Sizing, Selection, and Installation of Pressure-relieving Devices
Part I—Sizing and Selection

1 Scope

This standard applies to the sizing and selection of pressure-relief devices used in refineries, chemical facilities, and related industries for equipment that has a maximum allowable working pressure (MAWP) of 15 psig (103 kPag) or greater. The pressure-relief devices covered in this standard are intended to protect unfired pressure vessels and related equipment against overpressure from operating and fire contingencies.

This standard includes basic definitions and information about the operational characteristics and applications of various pressure-relief devices. It also includes sizing procedures and methods based on steady state flow of Newtonian fluids.

Pressure-relief devices protect a vessel against overpressure only; they do not protect against structural failure when the vessel is exposed to extremely high temperatures such as during a fire. See API 521 for information about appropriate ways of reducing pressure and restricting heat input.

Atmospheric and low-pressure storage tanks covered in API 2000 and pressure vessels used for the transportation of products in bulk or shipping containers are not within the scope of this standard.

The rules for overpressure protection of fired vessels are provided in ASME *BPVC*, Section I and are not within the scope of this standard.

2 Normative References

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

API Standard 520, *Sizing, Selection, and Installation of Pressure-relieving Devices in Refineries, Part II—Installation*

API Standard 521, *Pressure-relieving and Depressuring Systems*

API Standard 526, *Flanged Steel Pressure-relief Valves*

API Standard 527, *Seat Tightness of Pressure-relief Valves*

API Standard 2000, *Venting Atmospheric and Low-pressure Storage Tanks*

ASME *Boiler and Pressure Vessel Code (BPVC)* [1]*, Section I*: *Power Boilers*

ASME *Boiler and Pressure Vessel Code (BPVC), Section VIII: Pressure Vessels; Division 1*

ASME *Boiler and Pressure Vessel Code (BPVC)* Code Case 2203 [2], *Omission of Lifting Device Requirements for Pressure Relief Valves on Air, Water Over 140°F, or Steam Service*

ASME B31.1, *Power Piping*

ASME B31.3, *Process Piping*

ASME PTC 25, *Pressure Relief Devices*

[1] ASME International, 3 Park Avenue, New York, New York 10016, www.asme.org.
[2] Code Cases are temporary in nature and may not be acceptable in all jurisdictions. The user should verify the current applicability of the referenced Code Cases.

3 Terms and Definitions

For the purposes of this document, the following definitions apply. Many of the terms and definitions are taken from ASME PTC 25.

3.1
accumulation
The pressure increase over the MAWP of the vessel, expressed in pressure units or as a percentage of MAWP or design pressure. Maximum allowable accumulations are established by applicable codes for emergency operating and fire contingencies.

3.2
actual discharge area
actual orifice area
The area of a pressure-relief valve (PRV) is the minimum net area that determines the flow through a valve.

3.3
backpressure
The pressure that exists at the outlet of a pressure-relief device as a result of the pressure in the discharge system. Backpressure is the sum of the superimposed and built-up backpressures.

3.4
balanced pressure-relief valve
A spring-loaded pressure-relief valve that incorporates a bellows or other means for minimizing the effect of backpressure on the operational characteristics of the valve.

3.5
blowdown
The difference between the set pressure and the closing pressure of a pressure-relief valve, expressed as a percentage of the set pressure or in pressure units.

3.6
bore area
nozzle area
nozzle throat area
throat area
The minimum cross-sectional flow area of a nozzle in a pressure-relief valve.

3.7
built-up backpressure
The increase in pressure at the outlet of a pressure-relief device that develops as a result of flow after the pressure-relief device opens.

3.8
burst pressure
The value of the upstream static pressure minus the value of the downstream static pressure just prior to when the disk bursts. When the downstream pressure is atmospheric, the burst pressure is the upstream static gauge pressure.

3.9
burst pressure tolerance
The variation around the marked burst pressure at the specified disk temperature in which a rupture disk shall burst.

3.10
capacity
The rated capacity of steam, air, gas, or water as required by the applicable code.

3.11
chatter
The opening and closing of a PRV at a very high frequency (on the order of the natural frequency of the valve's spring mass system).

3.12
closing pressure
The value of decreasing inlet static pressure at which the valve disc reestablishes contact with the seat or at which lift becomes zero as determined by seeing, feeling, or hearing.

3.13
coefficient of discharge
The ratio of the mass flow rate in a valve to that of an ideal nozzle. The coefficient of discharge is used for calculating flow through a pressure-relief device.

3.14
cold differential test pressure
CDTP
The pressure at which a pressure-relief valve is adjusted to open on the test stand. The CDTP includes corrections for the service conditions of backpressure or temperature or both.

3.15
conventional pressure-relief valve
A spring-loaded pressure-relief valve whose operational characteristics are directly affected by changes in the backpressure.

3.16
curtain area
The area of the cylindrical or conical discharge opening between the seating surfaces above the nozzle seat created by the lift of the disc.

3.17
cycling
The relatively low frequency (a few cycles per second to a few seconds per cycle) opening and closing of a pressure-relief valve.

3.18
design pressure
Pressure, together with the design temperature, used to determine the minimum permissible thickness or physical characteristic of each vessel component as determined by the vessel design rules. The design pressure is selected by the user to provide a suitable margin above the most severe pressure expected during normal operation at a coincident temperature. It is the pressure specified on the purchase order. This pressure may be used in place of the MAWP in all cases where the MAWP has not been established. The design pressure is equal to or less than the MAWP.

3.19
effective coefficient of discharge
A nominal value used with an effective discharge area to calculate the relieving capacity of a pressure-relief valve per the preliminary sizing equations.

4 Pressure-relief Devices

4.1 General

This section describes the basic principles, operational characteristics, applications, and selection of pressure-relief devices used independently or in combination. These devices include spring-loaded and pilot-operated pressure-relief valves (PRVs), rupture disk devices, and other pressure-relief devices. These devices are described in the text and illustrated in Figure 1 through Figure 29.

4.2 Pressure-relief Valves (PRVs)

4.2.1 Spring-loaded PRVs

4.2.1.1 Conventional PRVs

4.2.1.1.1 A conventional PRV (see Figure 1 and Figure 4) is a self-actuated spring-loaded PRV that is designed to open at a predetermined pressure and protect a vessel or system from excess pressure by removing or relieving fluid from that vessel or system. The valve shown in Figure 4 is available in small sizes, commonly used for thermal relief valve applications. The basic elements of a spring-loaded PRV include an inlet nozzle connected to the vessel or system to be protected, a movable disc that controls flow through the nozzle, and a spring that controls the position of the disc. Under normal system operating conditions, the pressure at the inlet is below the set pressure and the disc is seated on the nozzle preventing flow through the nozzle.

4.2.1.1.2 Spring-loaded PRVs are referred to by a variety of terms, such as safety valves, relief valves, and safety relief valves. These terms have been traditionally applied to valves for gas/vapor service, liquid service, or multiservice applications, respectively. The more generic term, PRV, is used in the text and is applicable to all three.

4.2.1.1.3 The operation of a conventional spring-loaded PRV is based on a force balance (see Figure 5). The spring load is preset to equal the force exerted on the closed disc by the inlet fluid when the system pressure is at the set pressure of the valve. When the inlet pressure is below the set pressure, the disc remains seated on the nozzle in the closed position. When the inlet pressure exceeds set pressure, the pressure force on the disc overcomes the spring force and the valve opens. When the inlet pressure is reduced to the closing pressure, the valve recloses.

4.2.1.1.4 When the valve is closed during normal operation (see Figure 5, Item A) the system or vessel pressure acting against the disc surface (area A) is resisted by the spring force. As the system pressure approaches the set pressure of the valve, the seating force between the disc and the nozzle approaches zero.

4.2.1.1.5 In vapor or gas service, the valve may “simmer” before it will “pop.” When the vessel pressure closely approaches the set pressure, fluid will audibly move past the seating surfaces into the huddling chamber B. As a result of the restriction of flow between the disc holder and the adjusting ring, pressure builds up in the huddling chamber B (see Figure 5, Item B). Since pressure now acts over a larger area, an additional force, commonly referred to as the expansive force, is available to overcome the spring force. By changing the position of the adjusting ring, the opening in the annular orifice can be altered, thus controlling the pressure buildup in the huddling chamber B. This controlled pressure buildup in the huddling chamber will overcome the spring force causing the disc to move away from the nozzle seat, and the valve will pop open.

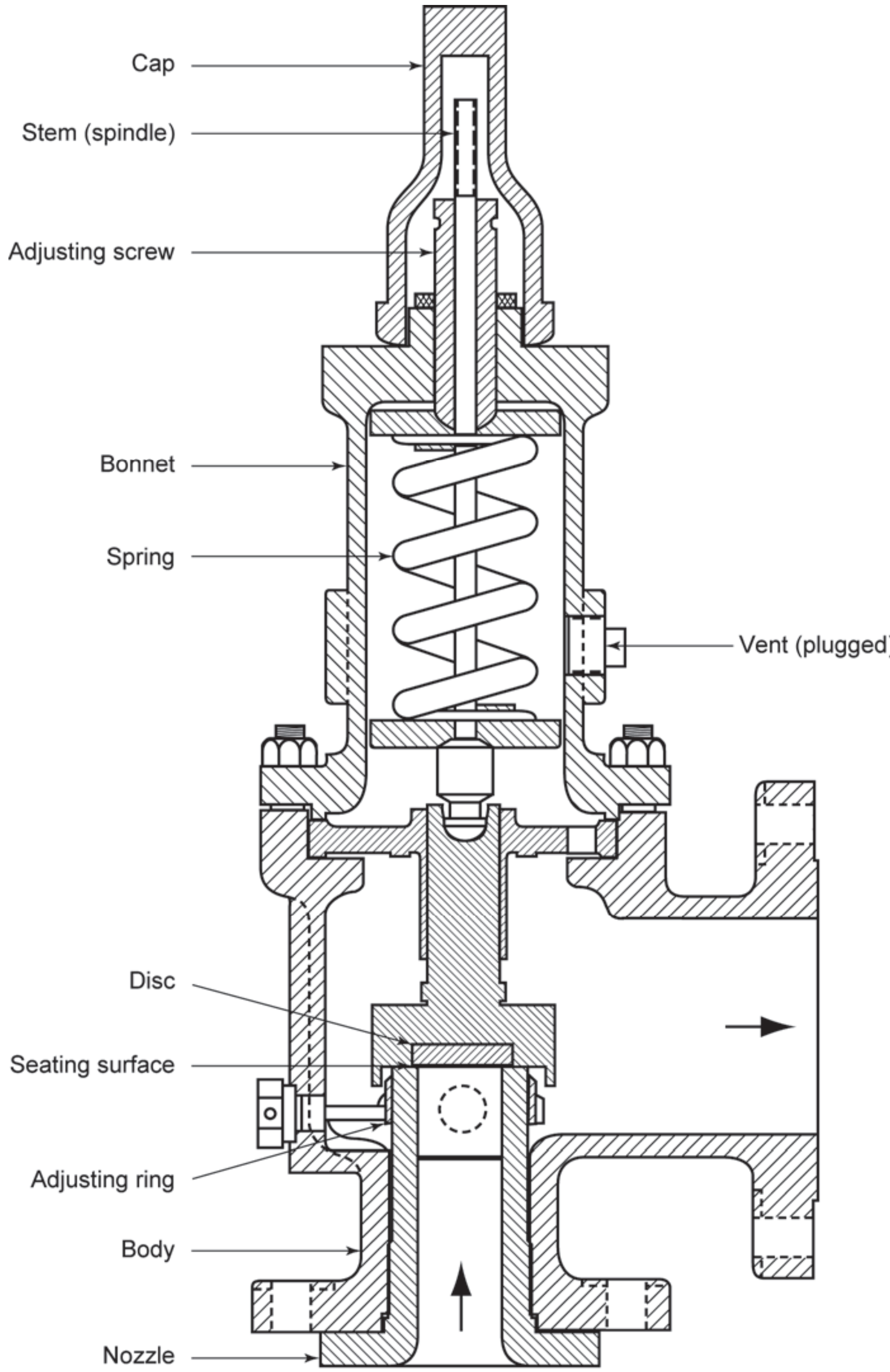

Figure 1—Conventional PRV with a Single Adjusting Ring for Blowdown Control